Cross-Document Reference Example

AMA3707 Real Analysis

This file demonstrates direct PDF references provided by `notes-common.sty`. Each reference contains enough information to link to the corresponding item in the deployed `lec01.pdf`; no reference index is required.

Example 1. definition natural number uses the document selected by `\xrefuse{lec01}`.

Example 2. definition integer explicitly names a document in this project.

Example 3. definition rational numbers explicitly names the project.

Example 4. Proposition link: rationals form a field.

Example 5. Convention link: zero is excluded from $\mathbb{N}_1$.

Example 6. remark number set inclusions uses the fully qualified server, project, document, and item.

Example 7. Multiple items can be linked together using definition integer, definition rational numbers.

Example 8. The starred form definition natural number prints without a link.

Example 9. Redirect link: an original ID that has been moved renders as a visible tombstone pointing to the new location. For example, definition converges to resolves through the redirect added when the predicate was renamed to `def:limit`.

Function Definition Style

Functions are defined in Lean-4 style: name, then type, then body, joined by $\equiv$. Multi-argument functions use the product type.

Example 1. Simple inline function:

$$f : \mathbb{R} \to \mathbb{R} \equiv x \mapsto x^2.$$

Example 2. Multi-argument function via product type:

$$g : \mathbb{R} \times \mathbb{R} \to \mathbb{R} \equiv (x, y) \mapsto x + y.$$

Example 3. Identity function (parametric type):

$$\mathrm{id} : A \to A \equiv x \mapsto x.$$

Example 4. Function with cases:

$$\mathrm{sign} : \mathbb{R} \to \{-1, 0, 1\} \equiv x \mapsto \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases}$$

Example 5. Restricted codomain (positive outputs only):

$$\mathrm{sq} : \mathbb{R} \to \mathbb{R}_{\geq 0} \equiv x \mapsto x^2.$$

Example 6. Partial operator (subset-type domain):

$$\text{half} : \{n \in \mathbb{Z} \mid n \text{ even}\} \to \mathbb{Z} \equiv n \mapsto n/2.$$

Example 7. Application of an anonymous mapsto (parens needed):

$$(x \mapsto x^2)(3) = 9.$$

Example 8. Nested application (direct, preferred over $\circ$ composition):

$$h \equiv x \mapsto f(g(x)), \qquad f(y) \equiv y + 1,\ g(y) \equiv y^2.$$